

Experiment 'p_2_4_e_distance' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_4_e_distance

Natural language statement: Write a method that takes two integers and then returns the distance (absolute value of the difference).

Method signature: p_2_4_e_distance(x: int, y: int) returns (result: int)

Ensures

- `result >= 0`
- `if x >= y then result == x - y else result == y - x`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes two integers and then returns the distance (absolute value of
→ the difference).

The signature should be:

`p_2_4_e_distance(x: int, y: int) returns (result: int)`

The method should respect the following contract:

`ensures result >= 0, ensures if x >= y then result == x - y else result == y - x`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXRVC41mAHgIbp6T33DOp1e3xaC

Model: gpt-5.5-2026-04-23

Created at: 1780484101

Finish reason: stop

Usage: 306 tokens in, and 74 tokens out

```
{  
  if x >= y {  
    result := x - y;  
  } else {  
    result := y - x;  
  }  
}
```

Final Program

```
method p_2_4_e_distance(x: int, y: int) returns (result: int)  
  ensures result >= 0  
  ensures if x >= y then result == x - y else result == y - x  
{  
  if x >= y {  
    result := x - y;  
  } else {  
    result := y - x;  
  }  
}
```

```
}  
}
```

Total Token Usage

Input tokens: 306

Output tokens: 74

Reasoning tokens: 23

Sum of ‘total tokens’: 380

Experiment Timings

Overall Experiment started at 1780484100565, ended at 1780484108038, lasting 7473ms (7.47 seconds)

Iteration #1 started at 1780484100565, ended at 1780484108033, lasting 7468ms (7.47 seconds)